# Yasmine Srioui

**Computer Science Engineering Student | AI/ML Engineering · LLM Systems · MLOps**
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## PROFILE

Final-year Computer Science Engineering student at the Higher National Engineering School of Tunis (ENSIT) with hands-on experience in LLM-powered applications, Graph RAG systems, and containerized ML services. Drawn to reliable, explainable, and human-centered AI. Seeking a 6-month end-of-studies internship (PFE) in France from January 2027.

## EDUCATION

**Higher National Engineering School of Tunis (ENSIT)** *Tunis, Tunisia*
Engineering cycle – Computer Science Engineering *September 2024 – Present*
**Preparatory Institute for Engineering Studies of Nabeul (IPEIN)** *Nabeul, Tunisia*
Preparatory cycle (Physics and Mathematics) *September 2023 – May 2024*

## EXPERIENCE

**Talan | Generative AI Internship** *(3rd place, Talan SummerCamp 2026)* Tunisia | July – August 2026
*Python, FastAPI, Mistral AI, Knowledge Graphs, Machine Learning*

- Co-developed **SANTAID**, an AI platform for French health insurance combining AI-assisted healthcare support with knowledge-graph-based fraud investigation.
- Developed the **insurer-side investigation dashboard**, connecting fraud-detection results to **Mistral-based generative AI** for explainable insights and human review.
- Contributed to an **agentic investigation system** that uses tool-based interactions with a knowledge graph to explore entities, relationships, timelines, and evidence.
- Worked in an agile, Scrum-inspired team with daily meetings and weekly deliveries presented to supervisors.

**AdsValley | Observation Internship** Tunisia | June 2025
*React, NestJS, PostgreSQL*

- Built a course management application in a professional software development environment.

## SELECTED PROJECTS

**CardioAgent-Ops: Containerized ML and Multi-Agent Platform** *In progress*
*Docker Compose, FastAPI, LangGraph, MLflow, PyTorch, ONNX, Qdrant, QLoRA, Pydantic, Pytest, GitHub Actions*

- Designing a **six-microservice architecture** for heart-sound analysis with Docker Compose: API gateway with guardrails, agent orchestrator, audio engine, vector database, LLM server, and MLflow tracking server.
- Benchmarking **five ML/DL models** (Decision Tree, SVM, CNN, ResNet transfer learning, Transformer-CNN) on merged PASCAL and PhysioNet data, selecting the best model by recall and exporting it to **ONNX**.
- Developing a **three-agent LangGraph workflow** (audio analysis, ESC-guideline RAG, QLoRA-fine-tuned LLM reasoning) with Pydantic-validated outputs, automated tests, and CI/CD with GitHub Actions.

**Echo Assistant: Fully Local Graph RAG System** *PFA2*
*Python, Flask, React, LanceDB, KuzuDB, Ollama, QLoRA, RAGAS*

- Built a local assistant combining vector retrieval, knowledge-graph traversal, and multi-hop reasoning, generating cited answers on hardware with 8 GB of VRAM.
- Fine-tuned **Qwen 2.5 7B** with QLoRA, evaluated on 448 RAGAS questions (faithfulness 0.79, answer relevancy 0.87).

**FreelanceHub: Freelance Marketplace Mobile Application** *Academic project*
*Flask, Ionic, MongoDB*

- Built a freelance marketplace with JWT authentication, role-based access control, real-time messaging, order management, and a RESTful API consumed by an Ionic mobile frontend.

**FlyWise: Flight Management Web Application** *PFA1*
*PHP, JavaScript, HTML, CSS, SQL*

- Developed a flight reservation and package management application, designing the relational database schema and implementing SQL queries.

## TECHNICAL SKILLS

**AI and LLM:** RAG, Graph RAG, Knowledge Graphs, LLM Fine-Tuning (QLoRA/PEFT), Agentic AI, LangGraph, Mistral, Qwen, Ollama, RAGAS, Prompt Engineering
**MLOps and Architecture:** Docker, Docker Compose, Microservices, MLflow, CI/CD (GitHub Actions), ONNX, Pytest, Git, Linux
**Machine Learning and Data:** Python, PyTorch, scikit-learn, Pandas, NumPy, Qdrant, LanceDB, KuzuDB
**Backend and Web:** FastAPI, Flask, REST APIs, Pydantic, Spring Boot, Java EE, React, Angular, NestJS, Ionic, PostgreSQL, MySQL, MongoDB
**Programming:** Python, C/C++, Java, JavaScript, SQL
**Languages:** Arabic (native), English (fluent), French (fluent), German (basic)

## ACTIVITIES AND CERTIFICATIONS

**Treasurer, ENSIT Geeks Club:** managed the club's budget and co-organized GeeksHack 5.0.
**Member:** IEEE Robotics and Automation Society (RAS), ENSIT
**Certification:** Cisco CCNA – Introduction to Networks